Quiz 7 Reflection

SCIE 001 Physics

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Contents

1. Neutron Beam Double-Slit

1.1. Specify Question

Question 8: Neutron Beam Double-Slit

You are reading a paper describing a neutron beam double-slit experiment. The experimental setup is as shown below in Figure 1, and the results are in Figure 2.

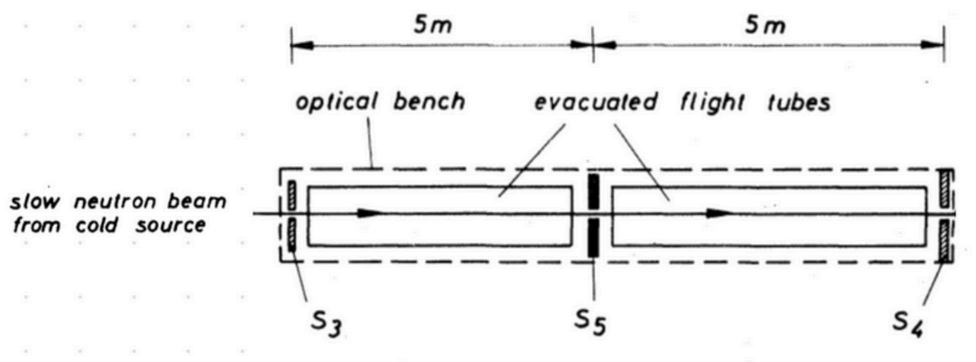

Figure 1: A collimated "slow" neutron beam enters the setup at **S3** and hits the double-slits at **S5**. The double-slits are $112\ \mu\mathrm{m}$ apart. The interference pattern is recorded at **S4**.

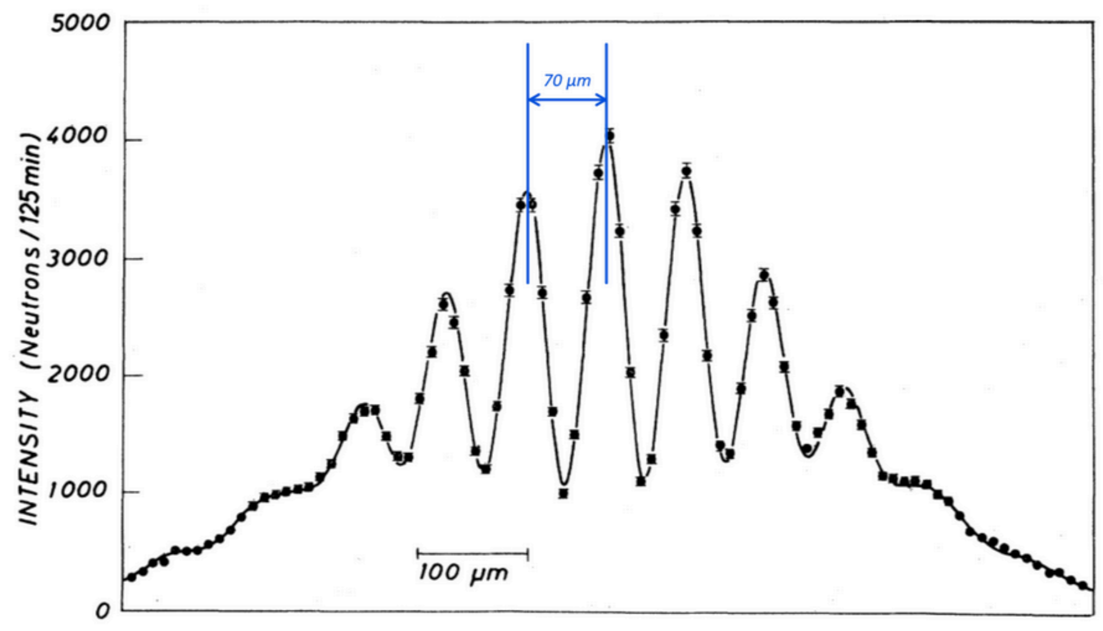

Figure 2: Interference pattern recorded at **S4**.

What was the approximate kinetic energy of these "slow" neutrons? Give your answer in $\mu\mathrm{eV}$ to the nearest integer.

▾ Useful constants

It's convenient to work in eV for this question. Recall that $hc = 1240\,\mathrm{eV}\cdot\mathrm{nm}$ and the mass of a neutron can be expressed as $939.6\,\mathrm{MeV}/c^2$. Alternatively, you can take the mass of the neutron as $1.675\times10^{-27}\,\mathrm{kg}$ and work in Joules; in which case, recall that $1\,\mathrm{eV} = 1.602\times10^{-19}\,\mathrm{J}$ and that $h = 6.626\times10^{-34}\,\mathrm{J\cdot s}$

$E_k =$ 40 $\mu\mathrm{eV}$ ✕ 0%

Correct answer

$E_k = 333\ \mu\mathrm{eV}$

1.2. Diagnosis Phase

During the test I forgot the conversion factor from μm to m and didn't realize that the result was in micro-electron volts, allowing you to cancel the meters and carry the micrometers through.

This caused me to panic, as with both ways of analyzing the question, you had to deal with micrometers.

Initially, I guessed $340eV$ as was the result of the initial calculation of the wavelength of the particle, but I changed that to $40eV$, as 340 looked like too much energy.

Words: 78

1.3. Correction Phase

I was incorrect in the units, carrying all the units forward and using momentum-energy equivalence we get:

$$E = \frac{p^2}{2m}$$

So we just need the momentum.

Using our small-angle approximation, we can reach an expression for fringe spacing in the double slit experiment (given to us on the slides):

$$\Delta y = y_{n+1} - y_n = \frac{\lambda L}{d}$$

Where d is the distance between the slits, and L is the length from the slits to the screen.

Solving for λ:

$$\frac{d\Delta y}{L} = \lambda$$

Using our equation for momentum, we can combine:

$$p = \frac{h}{\lambda}$$

$$p = \frac{h}{\frac{d\Delta y}{L}}$$

$$p = \frac{hL}{d\Delta y}$$

$$E = \frac{\left(\frac{hL}{d\Delta y}\right)^2}{2m}$$

$$E = \frac{h^2}{2md^2\Delta y^2}$$

To make the math easier, we can multiply both sides of the fraction by c^2.

$$E = \frac{h^2c^2}{2md^2\Delta y^2c^2}$$

$$E = \frac{(hc)^2}{2m(c^2)d^2\Delta y^2}$$

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